RI080

Typed Judgment, Deterministic Enforcement: A Consent-Gated Gateway Architecture for Inter-Agent Collaboration over Messaging Channels

Yizhang Cai[1] Jun Liu[1] Di Ma[2] Ke Yu[1] Ke Li[1]

[1]School of Artificial Intelligence, Beijing University of Posts and Telecommunications [2]Beijing Guoshu Zhilian Technology Co., Ltd.
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Introduction

Messaging channels deliver human events that are unstructured, duplicated, delayed, or replayed, while agent protocols expect authenticated endpoints, structured requests, and explicit task state. A gateway therefore has to decide whether a message requests work, which agent is eligible, whether consent is needed, and whether an event is a replay or an injection.
We separate bounded model judgment from deterministic enforcement: the model answers closed questions, and policy code owns thresholds and every effect.

Objectives

- Seven modules from an iMessage/Photon ingress to an ACPs/AIP v2.2 Leader–Partner path.
- Four invariants kept in code: authorization, identity, isolation, idempotency.
- A judgment plane of seven closed questions (J1–J7) that can only parameterize thresholds.
- Live measurement of routing, consent, and the deployed advisory path.

Judgment plane

The layer returns closed labels or probabilities only: it cannot emit free text or call a tool, and it only parameterizes policy thresholds.

J1 intent
J2 routing
J3 ambiguity
J4 consent
J5 replay
J6 retryability
J7 injection

Materials and Methods

- M1 ingress authentication; M2 caller/conversation isolation; M3 discovery and routing; M4 consent; M5 identity-bound execution; M6 state, idempotency and recovery; M7 delivery and telemetry.
- 130 generated requests through the real ingress, three deliveries each; paired same-code ablations on staging copies; consent trials on the deployed adapter; a 280-item judgment suite sampled three times.

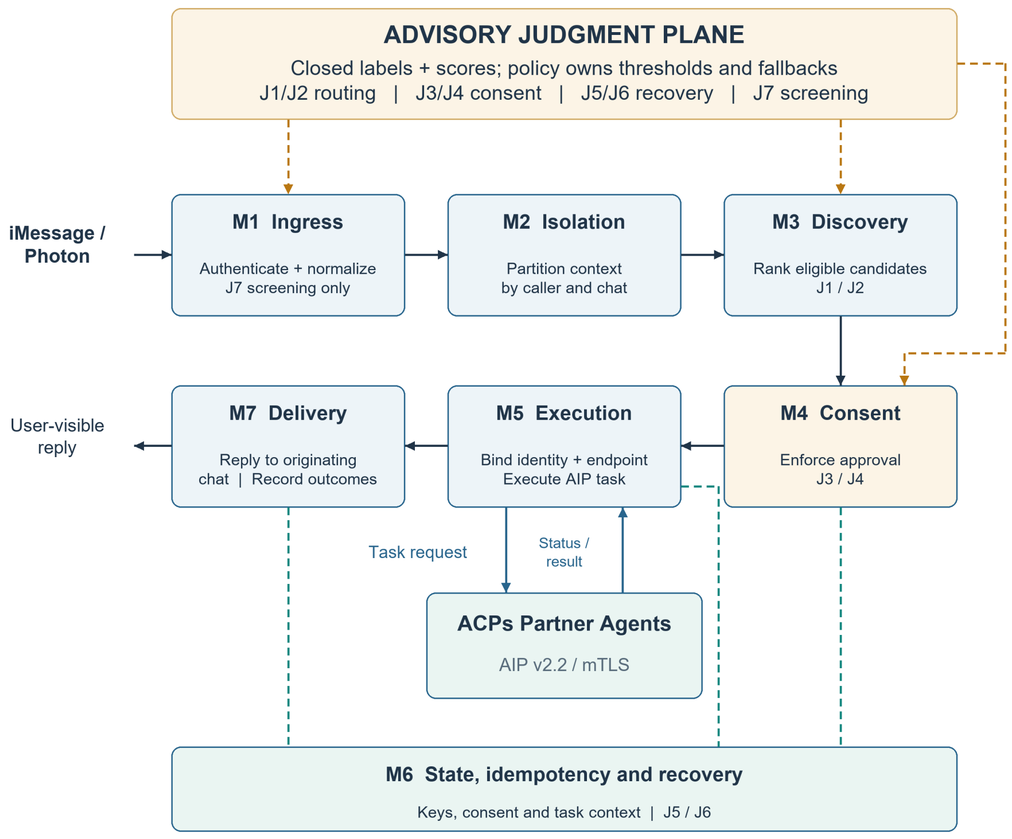


Figure 1. Seven modules, four invariants, and the advisory judgment plane.

Contributions

- A seven-module gateway that separates bounded judgment from deterministic enforcement in code.
- Four invariants enforced by the runtime rather than by prompts.
- A judgment plane of seven closed questions that cannot act on its own.
- A live evaluation of routing, consent and advisory judgment, with a released artifact.

Evaluation setup

Requests	130 generated, 3 deliveries each
Consent	30 trials + 10 redeliveries
Judgment suite	280 items, 3 runs
Injection stress	200 injected, 256 benign

At a glance

48/48/50
target recommended (of 50)

0/30 → 30/30
identity ablation

36/36
one action per consent

94.1% vs 83.9%
typed vs free text

Results

Routing. The generated target was recommended for 48/48/50 of 50 direct requests; 39/38/38 of 40 ambiguous requests received more than one agent; 0/40 out-of-scope requests recommended a study partner.

Ablation	Control	Removed
Identity binding	0/30	30/30
Caller isolation	0/30	30/30
Idempotency	30 exec.	60 exec.

Consent. No partner action before consent (0/40); every consent produced exactly one action (36/36).

	Typed	Free text
Accuracy	94.1%	83.9%
Brier	0.095	0.266
ECE	0.026	0.144

280 items, three runs, human-verified labels. The injection stress set (200 injected, 256 benign) misses none but flags about half of quoted mentions, which is why the injection score only parameterizes policy.

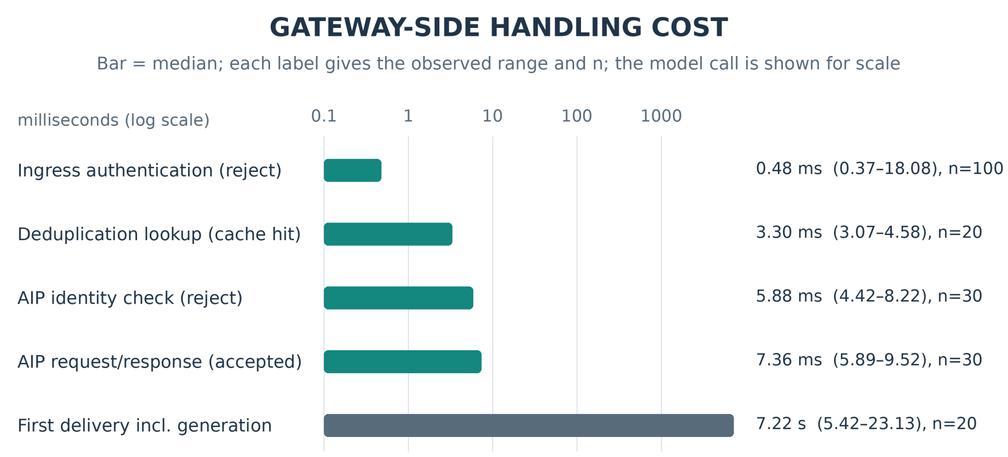


Figure 3. Median gateway-side handling cost per operation.

Processing flow

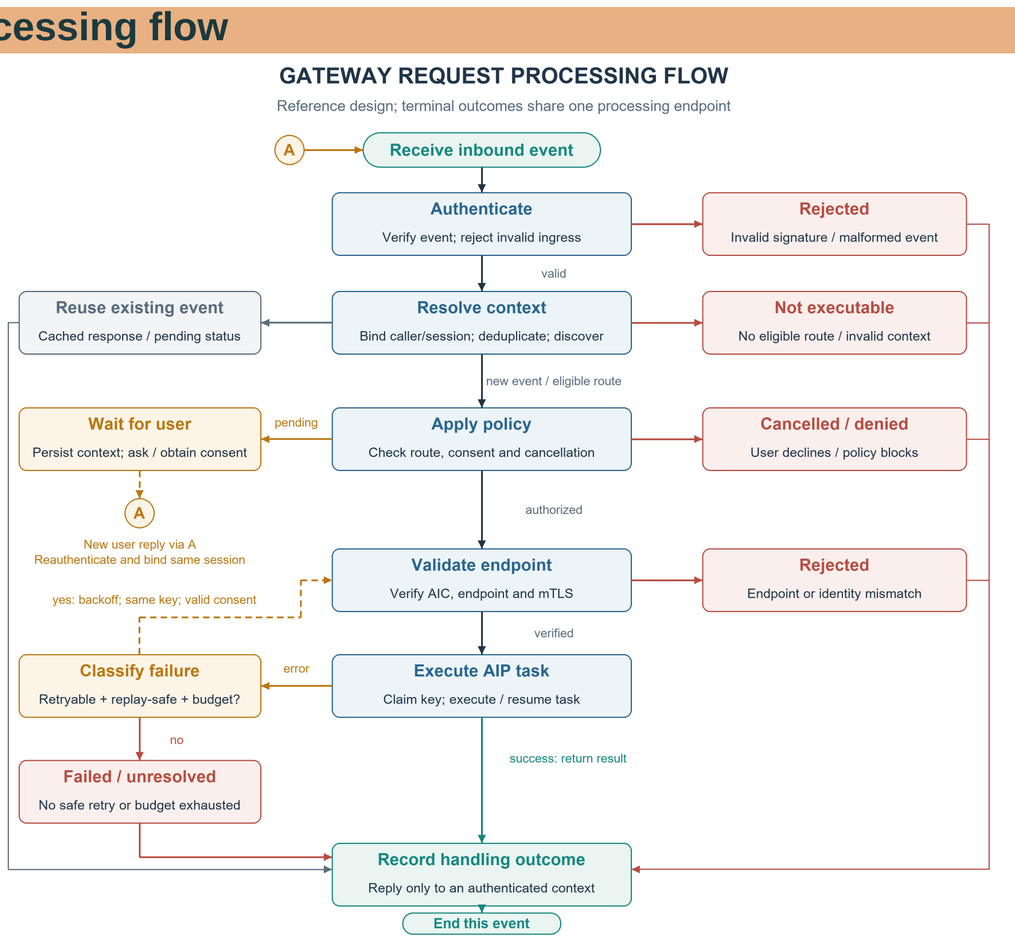


Figure 2. Deduplication precedes every model call; a pending consent re-enters at authentication; a failed identity or endpoint check never reaches execution.

Conclusion

The model advises; code controls identity, state, and every side effect. Typed answers are more accurate and better calibrated than free text, and consent gating admits exactly one partner action per accepted event.

References

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Contact

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Yizhang Cai, 2025111259@bupt.cn
Artifact: github.com/ankelas/consent-gated-gateway-artifact
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